LDA Homework 1 (2025-2026)

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Question 1

Data description

At baseline, all 1253 patients contribute BPRS, with a mean around 75; this increases to about 107 by the last visit, but by year 6 only 511 patients remain, the later years summary statistics such as mean are based on less than half of the original group. Over time, those who stay in the study tend to be younger, more often male, more likely to have a paid job, with higher ADL scores, and more often living at home rather than in a nursing home. ABPET0 also tends to be lower in later years, and TAUPET0 loses almost all variability from year 4 onward. Together this shows that dropout is not random, and long-term averages mostly reflect a healthier, more selected subset of patients.

Check if Data is Balanced

To confirm results from summary statistics describe above. We plot Figure 1 to show that patients with higher baseline BPRS are more likely to miss later visits, and the percentage missing rises to above 40% by year 6, so the number of patients contributing to the later means becomes much smaller. Simple averages by time point would then mainly reflect the patients who stay in the study and under-represent those who drop out (often the sicker ones), and can therefore give a biased trend. Because this pattern of missing causes an imbalanced data set, we use smoothing methods that are better suited for to investigate mean, variance and correlation structure over time.

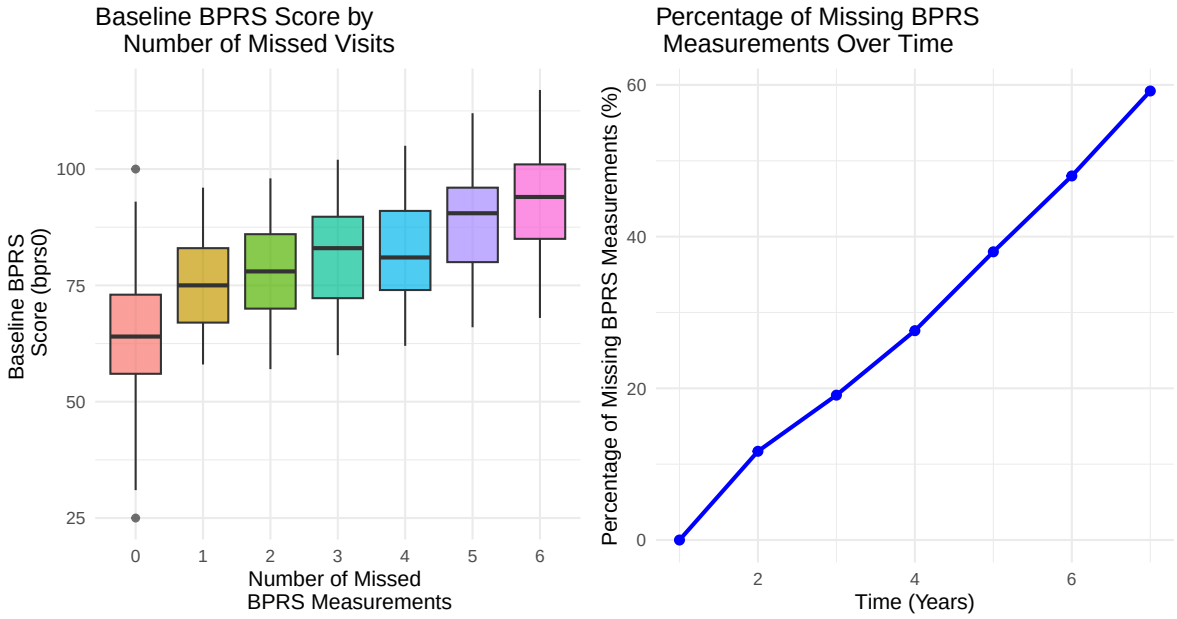


Figure 1: Missingness Analysis: Baseline BPRS by Number of Missed Visits (left) and Percentage Missing Over Time (right)

Average loess smoothing

Categorical Variables

To explore potential subgroup differences and to see how various variables are associated with the evolution of BPRS, we create LOESS smoothed plots (Figure 2) of BPRS over time stratified by categorical variables: sex, education, job status, living situation, and clinical center. Across these variables, BPRS increases over time. Most subgroup curves are roughly parallel, showing main effects without interactions for variables like sex, job status, and living situation. The four education categories overlap almost completely and there is no difference in trend. This means there is unlikely a significant effect of education on BPRS trajectory.

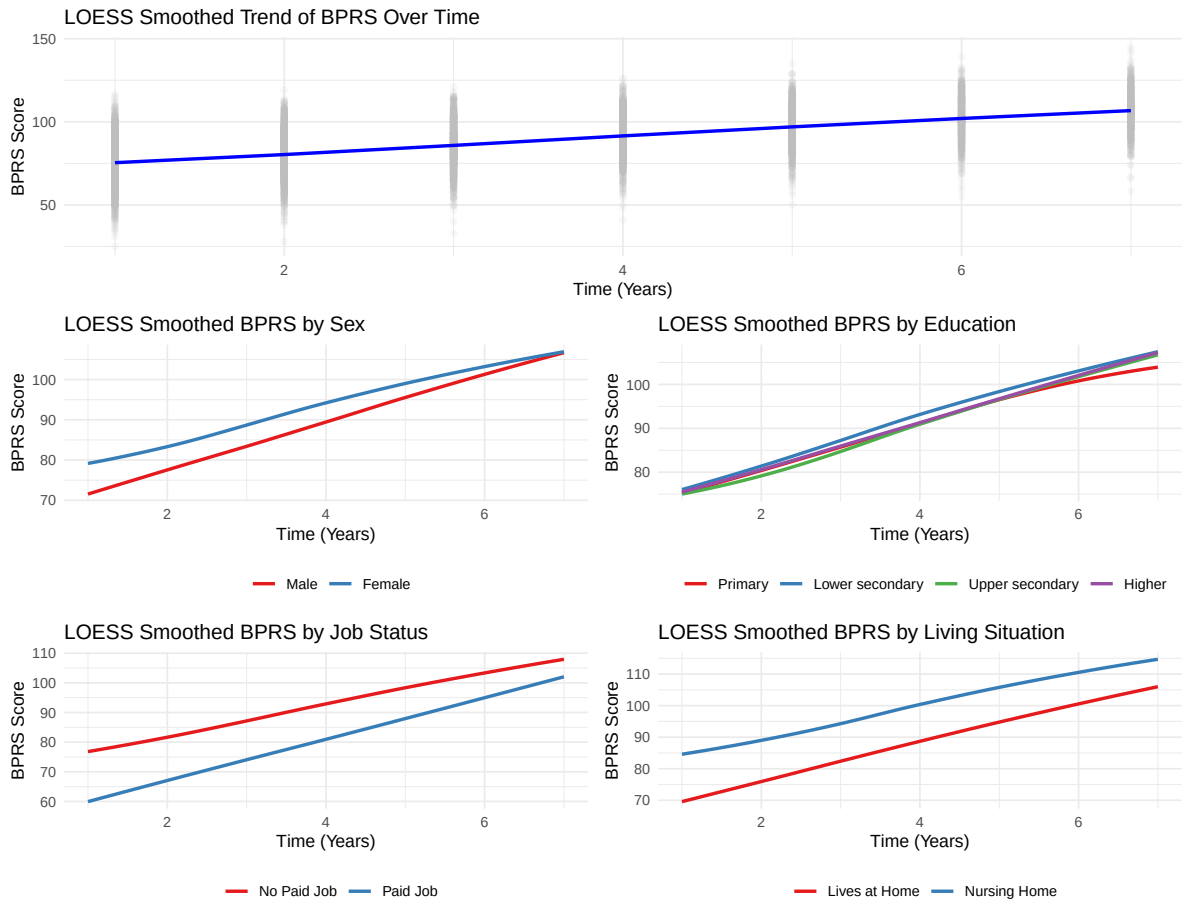


Figure 2: LOESS Smoothed BPRS Over Time by Subgroups. Overall average (top), sex, education, job status, living situation, and clinical center.

Continuous Variables

To explore continuous predictors, we plotted (Figure 3) LOESS-smoothed BPRS against age, BMI and baseline CDR-SB (cdrsb0) and abept0 using separate curves for each follow-up year. Higher age is associated with higher BPRS at every visit. BMI shows a curved pattern that peaks in the mid-20s, suggesting that a non-linear term would describe it better than a straight line. Baseline CDR-SB shows only a weak and slightly curved association with BPRS. The LOESS plot of BPRS versus ABPET0 by visit year shows an unstable curved pattern, with peak at year 3 that is not repeated at other visits, which might be because of high missingness of this variable in follow up years

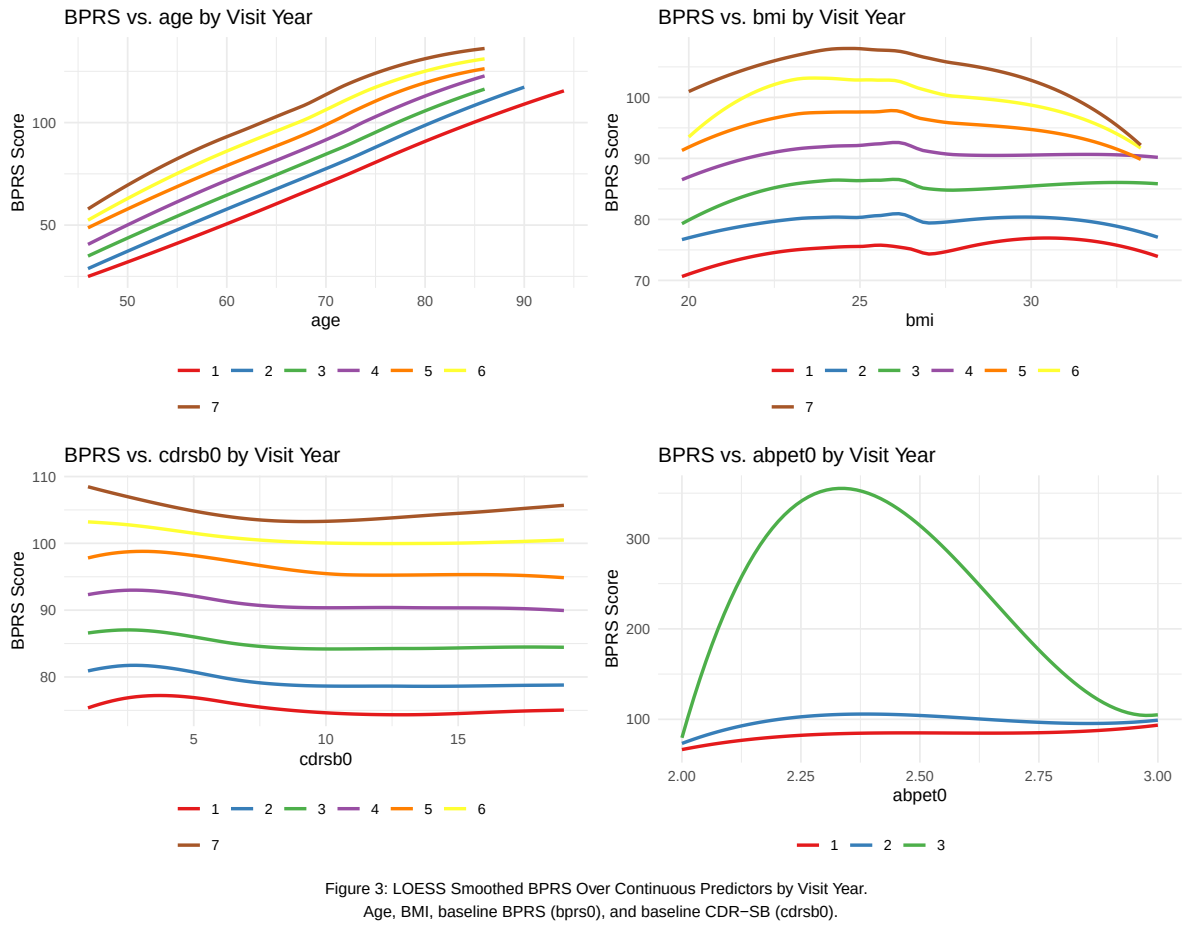


Figure 3: LOESS Smoothed BPRS Over Continuous Predictors by Visit Year. Age, BMI, baseline BPRS (bprs0), and baseline CDR-SB (cdrsb0).

Trial Centers Plot

To check whether clinical center (trial) affects the mean structure, we plot(Figure 4) LOESS-smoothed BPRS over time separately by trial centre. We see clear separate curves between centers, indicating that trial should be included as a random effect in the model to account for between center variability.

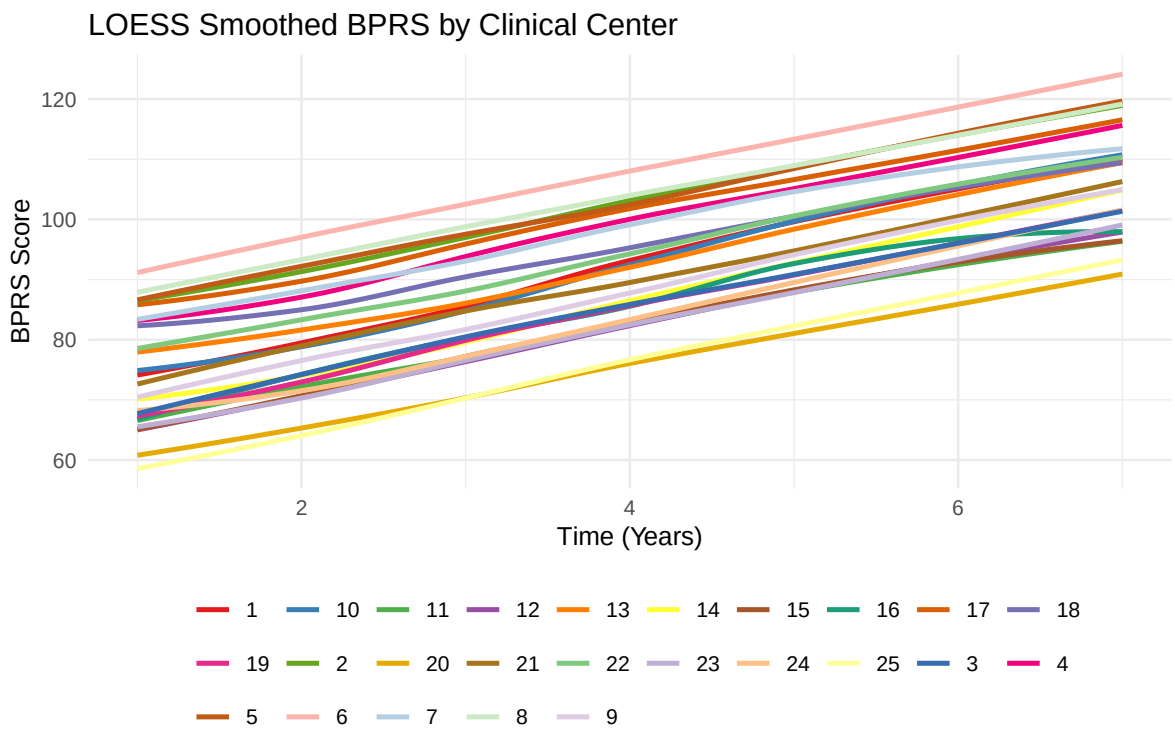


Figure 4: LOESS Smoothed BPRS Over Time by Clinical Center.

Variance & Correlation Structure

The smoothed squared residuals shows a flat residual curve, we dont observe a heteroscedastic pattern, meaning the spread of patients BPRS's readings trajectory around there expected readings doesn't change a lot. The semi-variogram also rises with the time gap between measurements, showing that observations from the same patient are more similar when close together and become less similar as the gap widens. These pattern indicates a within-patient correlation that weakens with distance in time therefore, the models we fit need to allow correlation structure that decays over time. The plots are shown in figure 5 below.

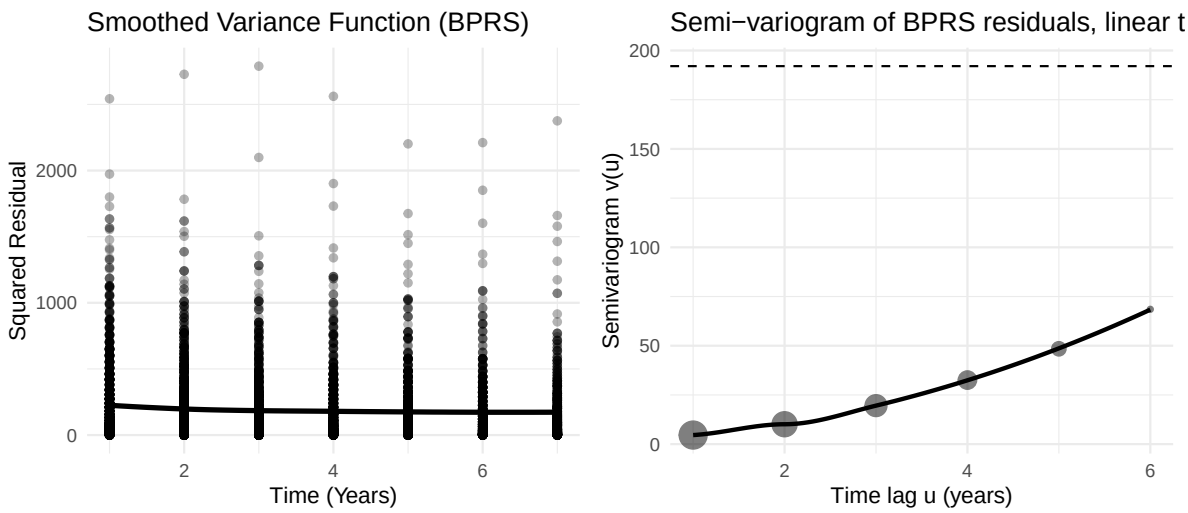


Figure 5: Smoothed Variance Function (left) and Semi-variogram (right) of BPRS Residuals.

Question 1: Conclusions and modeling implications

- Mean structure: BPRS rises steadily over time. Subgroup curves are mostly parallel with level shifts by sex, job, living situation, and clear vertical shifts by clinical center; education shows little separation.
- For continuous variables age, showed a straight line pattern while BMI showed non linear relationship with BPRS.
- Variance structure: The smoothed squared-residual function is roughly flat across visits (no visual heteroscedasticity observed). We will fit a model with assumed heterogeneous variance and compare this with a constant variance model using AIC.
- Correlation structure: The semivariogram increases with time lag, indicating strong within-patient correlation at short lags that decays as visits are farther apart.
- Missingness: Dropout increases over time and is more common for patients with higher baseline BPRS, so simple time-point averages would be biased
- Modeling implications: For population-average inference, we will fit repeated measures model with unstructured covariance structure and compare with AR(1) correlation
- For subject-specific inference, we will fit a mixed model with patient random intercept and slope and include a random intercept for clinical center, because there is a clear clinical site differences
- we don't recommend using taupet 0 as a covariate in longitudinal models as it has zero variance from year 3 onwards, for the participants who's BPRS measurement were taken at year 3 and beyond. Hence it will not show how taupet0 affects BPRS over the years

Question 2

Individual profiles & per-patient slopes

To visualise heterogeneity in longitudinal evolution, BPRS trajectories of 40 random patients were plotted (Figure 6). These individual profiles show variation in baseline symptom severity, together with increasing trends over time. The profiles also illustrate the unbalanced nature of the data, as subjects contribute different numbers of observations. These observations help justify the choice of summary statistics for longitudinal data.

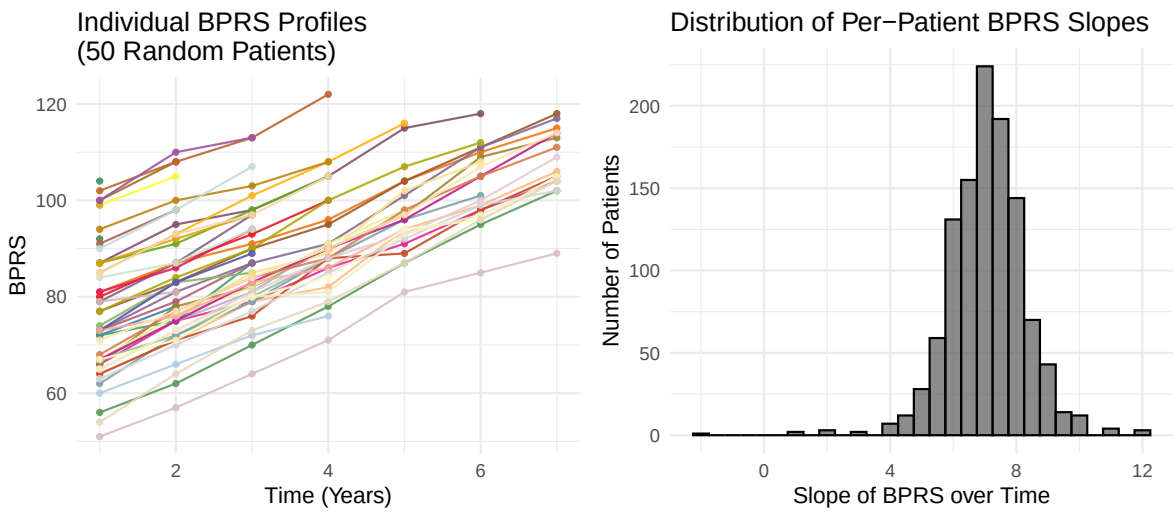


Figure 6: Individual BPRS Profiles (left) and Distribution of Per-Patient Slopes (right).

Increment Analysis

Increment analysis examines the year-to-year changes in BPRS scores. We calculate the increments (differences) between consecutive years for each patient and then summarize these increments across all patients at each time point.

Motivation

This approach is motivated by the goal of understanding how BPRS evolves over time and by the amount of baseline heterogeneity observed in the data; differencing removes these baseline differences and focuses on the rate of change. Increments were estimated by sorting each patient's measurements by time and subtracting $BPRS_{t-1}$ from $BPRS_t$.

Assumptions and limitations

This method assumes that the measurement intervals are comparable across subjects and that the change between two visits is meaningful even in the presence of unbalanced follow-up. It has the advantage of being model-free and directly interpretable in terms of annual worsening, although it uses only patients with consecutive measurements and can be influenced by measurement error at a single visit.

Results

The resulting plot(Figure 7) shows that, the mean year-to-year increment in BPRS is roughly constant across all follow-up years, with an average change of about seven points per year. The error bars represent the standard deviation of these increments, indicating that the variability in annual change is also relatively constant over time. This consistency suggests that, on average, patients deteriorate at a steady rate during the follow-up.

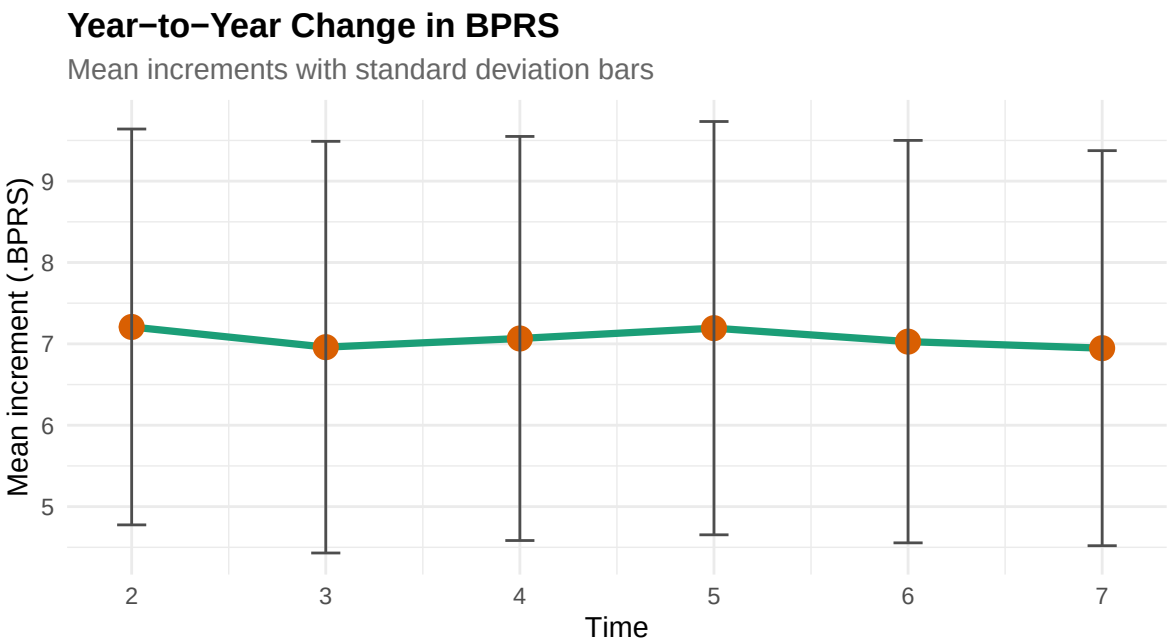


Figure 7: Year-to-Year Change in BPRS. Mean increments with standard deviation bars.

Separate analyses for each follow-up year

For each follow-up year $t \in \{1, \dots, 6\}$, we fit an ordinary linear model relating the BPRS score at year t to baseline covariates:

Table 1: Separate Linear Models for Each Follow-up Year

covariate	Year0	Year1	Year2	Year3	Year4	Year5	Year6
Intercept	-122.72 [-126.28, -119.16]	-119.78 [-126.03, -113.52]	-105.57 [-120.15, -90.99]	-104.92 [-111.33, -98.51]	-101.46 [-109.63, -93.29]	-95.55 [-106.07, -85.02]	-87.73 [-101.67, -73.79]
Baseline Amyloid PET	0.34 [-0.07, 0.74]	0.20 [-0.29, 0.69]	0.03 [-0.61, 0.68]	-0.13 [-0.98, 0.71]	-0.03 [-1.18, 1.13]	-0.10 [-1.74, 1.54]	-1.11 [-3.75, 1.54]
Activities of Daily Living Score	2.01 [1.92, 2.11]	2.17 [2.05, 2.29]	2.25 [2.09, 2.41]	2.11 [1.91, 2.32]	2.26 [2.00, 2.52]	2.37 [2.02, 2.71]	2.42 [1.96, 2.87]
Age (years)	2.36 [2.33, 2.40]	2.41 [2.36, 2.45]	2.42 [2.37, 2.48]	2.41 [2.34, 2.48]	2.45 [2.36, 2.54]	2.48 [2.37, 2.58]	2.50 [2.37, 2.64]
Body Mass Index	0.49 [0.44, 0.55]	0.52 [0.45, 0.58]	0.51 [0.42, 0.59]	0.54 [0.43, 0.65]	0.56 [0.42, 0.70]	0.51 [0.33, 0.69]	0.47 [0.23, 0.71]
Baseline CDR-SB Score	-0.02 [-0.04, -0.01]	-0.04 [-0.06, -0.02]	-0.06 [-0.08, -0.04]	-0.08 [-0.11, -0.05]	-0.13 [-0.17, -0.09]	-0.15 [-0.20, -0.10]	-0.17 [-0.23, -0.10]
Paid Job vs No Paid Job	-18.33 [-19.09, -17.57]	-19.24 [-20.17, -18.30]	-19.77 [-20.97, -18.56]	-19.01 [-20.54, -17.47]	-20.19 [-22.13, -18.25]	-20.86 [-23.30, -18.41]	-21.07 [-24.18, -17.96]
Sex: Female vs Male	-0.28 [-0.51, -0.05]	-0.40 [-0.68, -0.12]	-0.18 [-0.54, 0.18]	-0.22 [-0.68, 0.25]	-0.34 [-0.94, 0.26]	-0.37 [-1.15, 0.41]	-0.57 [-1.64, 0.49]
Baseline Tau PET	0.43 [-0.57, 1.43]	0.37 [-2.34, 3.09]	-4.06 [-11.47, 3.35]	NA	NA	NA	NA
Lives in Nursing Home vs Home	1.69 [1.43, 1.94]	1.94 [1.63, 2.25]	2.10 [1.69, 2.51]	2.24 [1.69, 2.80]	1.94 [1.18, 2.71]	1.43 [0.34, 2.52]	2.16 [0.42, 3.90]

$$
\begin{aligned}
extBPRS_{it} = \alpha_{0t} &+ \alpha_{1t} \cdot \text{sex}_i + \alpha_{2t} \cdot \text{age}_i + \alpha_{3t} \cdot \text{bmi}_i + \alpha_{4t} \cdot \text{job}_i + \alpha_{5t} \cdot \text{adl}_i \\
&+ \alpha_{6t} \cdot \text{wzc}_i + \alpha_{7t} \cdot \text{cdrsb0}_i + \alpha_{8t} \cdot \text{abpet0}_i + \alpha_{9t} \cdot \text{taupet0}_i + \varepsilon_{it}.
\end{aligned}
$$

where, for each fixed year t:

- α_{0t} is the expected BPRS at year t for the reference patient (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- α_{1t} is the difference at year t for Female vs Male.
- α_{2t} is the change at year t per 1-year increase in age.
- α_{3t} is the change at year t per 1-unit increase in BMI.
- α_{4t} is the difference at year t for Paid Job vs No Paid Job.
- α_{5t} is the change at year t per 1-unit increase in ADL.
- α_{6t} is the difference at year t for Nursing Home vs Lives at Home.
- α_{7t} is the change at year t per 1-unit increase in baseline CDR-SB (cdrsb0).
- α_{8t} is the change at year t per 1-unit increase in baseline Amyloid PET (abpet0).
- α_{9t} is the change at year $t" per 1-unit increase in baseline Tau PET (taupet0).
- ε_{it} is the error term at year t.

Motivation

This approach treats each follow-up visit as a cross-sectional outcome and regresses the BPRS score at that year on baseline demographic variables and clinical/bio-marker measures. While a full multivariate model may be difficult to specify, analysing each time point separately provides a simple way to study whether the effects of baseline covariates vary over time; making this technique appropriate. Each model uses all patients who remained under observation at that specific time, making the method suitable with unbalanced longitudinal designs.

Assumptions and limitations

The estimates were obtained using standard least squares for each year, assuming independent errors within each year-specific regression, and assuming that baseline covariates affect follow-up BPRS levels cumulatively. A limitation is that subjects with missing biomarker values are excluded for the affected years, and predictors that show no variation within a given year (e.g., taupet0 from year 3 onwards) cannot be included in that year's model. This leads to different sample sizes and covariate sets across years, complicating direct comparisons of effects over time.

Results

Across all years, several patterns emerged. Age, BMI, and ADL showed consistent positive associations with BPRS at each follow-up, and the magnitude of these effects remained constant over time. Patients living in a nursing home at baseline consistently had higher BPRS scores. Employment status also showed a persistent effect, with patients in paid jobs having lower BPRS scores at every follow-up point. CDR-SB at baseline had a small but consistent negative association with later BPRS. By contrast, ABPET and TAUPET displayed large uncertainty and no consistent effect across years.

Conclusion

Across both analyses, patients show a steady increase in psychiatric symptoms, with BPRS scores rising by roughly seven points each year on average. This rate of worsening symptoms remains stable throughout the six-year follow-up, with no evidence that it accelerates or slows down as the disease progresses.

Patients who begin with greater functional impairement, older age, or who reside in a nursing home demonstrate higher BPRS scores throughout follow-up. In contrast, employed individuals at baseline tend to have consistently lower symptom severity over time. Baseline amyloid and tau biomarkers do not show consistent associations with BPRS trajectories.

These findings suggest that the baseline factors associated with higher psychiatric severity remain consistent at all later points in follow-up. The effect sizes do not appear to change considerably over time, suggesting that the influence of these baseline characteristics on BPRS levels is stable rather than time-dependent. This supports the idea that demographic and functional status factors at baseline could be useful markers for identifying patients who are likely to experience more severe psychiatric symptoms throughout their disease course. This information can help anticipate the likely trajectory of symptom worsening and plan follow-up and supportive care accordingly.

Question 3

Motivation Repeated-Measures Model

In Question 2, we analysed each year separately and presented yearly results (Table 1). While this approach is simple, it has limitations. Running separate models for each year means repeating the same hypothesis test multiple times, which increases the risk of Type I error. It also reduces statistical power, because each model uses only one year of data instead of leveraging all observations together. Additionally, separate analyses do not address the main research question: how BPRS scores changes over time and how baseline characteristics influence that change. They also ignore the fact that repeated measurements from the same patient are correlated, which can lead to misleading conclusions.

To overcome these limitations when estimating the population effects, we use a repeated-measures multivariate model. A normal multiple linear model assumes all observations are independent, which is not realistic here because the same patient BPRS measurements are taken yearly up to year 6. The repeated measures model correctly handles the correlation within a person, by the specification of a covariance structure for the within-patient errors.

We start from a saturated repeated measures model where time, all baseline characteristics, and their interactions with time are included. This gives us a way to assess how the variable affect both the starting BPRS level and the change over time, so that we do not miss important effects when we later simplify the model. For the repeated measurements within a patient, we allow a very flexible pattern for how BPRS scores at different visits are correlated to each other, without forcing a specific correlation pattern in advance. We do not use this saturated model to report final clinical results, we use it as a screening model and then simplify the mean structure.

We start from the following marginal mean model:

$$
\begin{aligned}
Y_{ij} = \beta_0 &+ \beta_1 \,\mathrm{time}_j + \beta_2 \,\mathrm{edu}_i + \beta_3 \,\mathrm{sex}_i + \beta_4 \,\mathrm{age}_i + \beta_5 \,\mathrm{bmi}_i \\
&+ \beta_6 \,\mathrm{job}_i + \beta_7 \,\mathrm{adl}_i + \beta_8 \,\mathrm{wzc}_i + \beta_9 \,\mathrm{cdrsb0}_i + \beta_{10} \,\mathrm{abpet0}_i \\
&+ \beta_{11}(\mathrm{time}_j \cdot \mathrm{sex}_i) + \beta_{12}(\mathrm{time}_j \cdot \mathrm{age}_i) + \beta_{13}(\mathrm{time}_j \cdot \mathrm{bmi}_i) + \beta_{14}(\mathrm{time}_j \cdot \mathrm{job}_i) \\
&+ \beta_{15}(\mathrm{time}_j \cdot \mathrm{adl}_i) + \beta_{16}(\mathrm{time}_j \cdot \mathrm{wzc}_i) + \beta_{17}(\mathrm{time}_j \cdot \mathrm{cdrsb0}_i) + \beta_{18}(\mathrm{time}_j \cdot \mathrm{abpet0}_i) + \varepsilon_{ij}
\end{aligned}
$$

- Y_{ij} is the BPRS score for patient i at time j.
- β_0 is the mean BPRS at time 0 for the reference patient (male, primary education, no paid job, living at home, with baseline continuous covariates at their reference value).
- β_1 is the average yearly change in BPRS for this reference patient.

- The main-effect terms $\beta_2, \dots, \beta_{10}$ describe how the BPRS level at time 0 shifts with each baseline covariate.
- The interaction terms $\beta_{11}, \dots, \beta_{18}$ multiply time and show how the yearly change in BPRS depends on baseline covariates. For example, β_{17} for $\text{time}_j \cdot \text{cdrsb0}_i$ is the change in the annual BPRS slope per 1-unit higher baseline CDR-SB.

Categorical predictors (sex, edu, job, wzc) are encoded as factors in R. Their coefficients represent contrasts relative to the reference levels shown above; for education, the single symbol β_2 denotes a vector of three contrasts (levels 2–4 versus Primary).

The within-patient errors $e_i = (e_{i1}, \dots, e_{in_i})^T$ are assumed to follow a multivariate normal distribution, $e_i \sim N(0, \Sigma_i)$, where Σ_i is a covariance matrix that accounts for correlation and heteroscedasticity. In this model, we use an unstructured correlation matrix (`corSymm`) and allow for different variances at each time point (`varIdent`).

Note: taupet0 terms (main effect and time interaction) were excluded from this model due to near-zero variability at later follow-up years. Which prevents stable estimation of their effects in a marginal mean model.

The complex repeated-measures model that includes time, all baseline characteristics, and all their interactions with time, while allowing for unstructured covariance structure. This complex model(Table 7) confirms that BPRS increases over time and that age, BMI, ADL, job status, and living in a nursing home are all associated to BPRS levels. However, most of interactions between time and other covariates are not significant, which tells us we can safely move to a simpler mean structure without losing important clinical information.

Simplified Mean Structure

For this step we fit a repeated-measures model with a simplified mean structure but the same unstructured covariance as before. In this model, time and all main baseline characteristics except education remain in the mean structure, and only one time interaction (time with baseline CDR-SB) is kept. The estimated correlation matrix shows a positive correlation between repeated BPRS scores within a patient, especially for visits that are closer in time, and the visit-specific standard deviations increase from baseline to later years, contradicting with our earlier variance plots showing homoscedastic variance. We will use this reduced-mean, unstructured covariance model as the starting point when we compare with simpler covariance structures for instance AR(1)

We use the following simplified mean structure:

Table 2: Simplfied mean structiure Unstructured Covariance Results

term	estimate	std.error	statistic	p.value
(Intercept)	-131.59 [-134.11, -129.08]	1.28	-102.43	<0.01
time	7.22 [7.14, 7.30]	0.04	179.38	<0.01
sexFemale	-0.27 [-0.46, -0.07]	0.10	-2.72	<0.01
age	2.39 [2.37, 2.42]	0.01	212.64	<0.01
bmi	0.51 [0.47, 0.55]	0.02	22.80	<0.01
jobPaid Job	-18.69 [-19.30, -18.08]	0.31	-59.80	<0.01
adl	2.08 [2.01, 2.16]	0.04	53.17	<0.01
wzcNursing Home	1.84 [1.62, 2.05]	0.11	16.86	<0.01
cdrsb0	0.00 [-0.01, 0.02]	0.01	0.35	0.73
time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.29	<0.01

$$
\begin{aligned}
Y_{ij} = \beta_0 &+ \beta_1 \,\mathrm{time}_j + \beta_2 \,\mathrm{sex}_i + \beta_3 \,\mathrm{age}_i + \beta_4 \,\mathrm{bmi}_i + \beta_5 \,\mathrm{job}_i \\
&+ \beta_6 \,\mathrm{adl}_i + \beta_7 \,\mathrm{wzc}_i + \beta_8 \,\mathrm{cdrsb0}_i + \beta_9 \,\mathrm{taupet0}_i + \beta_{10} \,\mathrm{abpet0}_i \\
&+ \beta_{11} \,(\mathrm{time}_j \cdot \mathrm{cdrsb0}_i),
\end{aligned}
$$

- Y_{ij} is the BPRS score for patient i at time j.
- β_0 is the mean BPRS at baseline for the reference patient (male, primary education, no paid job, living at home, with baseline continuous covariates at their reference value).
- β_1 is the average yearly change in BPRS for this reference patient.
- The main-effect terms $\beta_2, \dots, \beta_{10}$ describe how the BPRS level at time 0 shifts with each baseline covariate.
- The interaction terms $\beta_{11}, \dots, \beta_{18}$ multiply time and show how the yearly change in BPRS depends on baseline covariates; for example, β_{17} for $\mathrm{time}_j \cdot \mathrm{cdrsb0}_i$ is the change in the annual BPRS slope per 1-unit higher baseline CDR-SB.

The covariance structure remains the same as in the full model.

LR test between full and reduced mean structure models

The likelihood ratio test compares the complex mean structure model (with all time–covariate interactions) to the reduced model (only time $\times$ CDR-SB). The test statistic is 13.78 with a p-value of 0.25, which means there is no evidence that the extra interaction terms improve the fit. This shows the simpler mean structure explains the data just as well as the saturated one, while using fewer parameters, so we will keep the reduced model as our working mean structure.

Reduced Covariance Structure

As a sensitivity check, we refitted the reduced mean model with a simpler AR(1) covariance, where each visit is mainly correlated with the previous one. The likelihood ratio test clearly favoured the unstructured covariance ($L.Ratio \approx 1233$, $p < 0.0001$) and AIC/BIC were much worse for AR(1). This confirms the lecture notes, with our unbalanced follow-up and dropout, simple covariance patterns such as AR(1) are too restrictive, so we keep the unstructured covariance.

Residual Diagnostics for Reduced Mean Structure Model

The residual diagnostics(Figure 8) for the reduced mean model are acceptable. The QQ-plot shows residuals close to the straight line, the normality assumption is satisfied, and the residuals vs fitted plot gives a symmetric points around zero without clear structure, suggesting the mean structure is well specified. Residuals by time are centred near zero with similar spread across visits, indicating variance homogeneity over time. Overall, these diagnostics support the adequacy of the reduced mean structure model with unstructured covariance for our data.

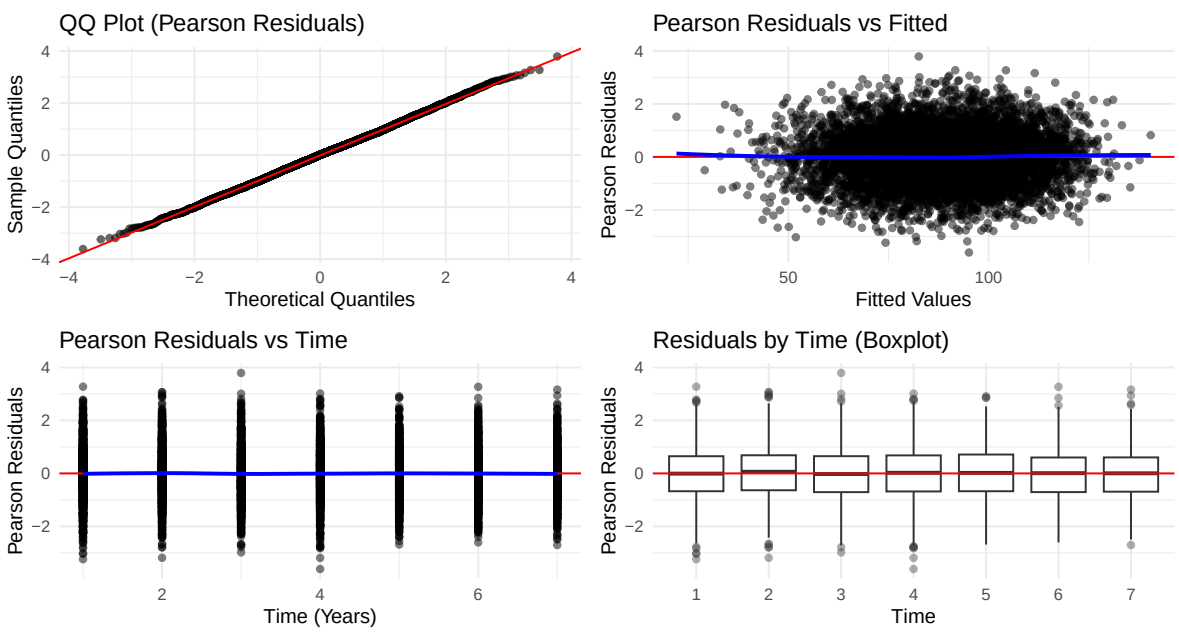


Figure 8: GLS Reduced Mean Structure Diagnostics: QQ plot, Residuals vs Fitted, Residuals vs Time.

Discussion of Question 3 Results

For Question 3 (Table 2), our final repeated-measures model uses a reduced mean structure with an unstructured covariance. On the mean side, BPRS rises on average by about 7.2 points per year (time effect $\hat{\beta}_{\text{time}} \approx 7.22$). After adjustment, older age, higher BMI, worse ADL, not having a paid job, and living in a nursing home are all linked to higher BPRS: for example, a paid job is associated with about 18.7 points lower BPRS (jobPaid Job ≈ -18.69), each unit of ADL with about 2.1 points higher BPRS (ADL ≈ 2.08), BMI with about 0.5 points higher BPRS (BMI ≈ 0.51), and living in a nursing home with about 1.8 points higher BPRS (WZC ≈ 1.84). Women have slightly lower scores than men (sexFemale ≈ -0.27). The time–CDRSB interaction is clearly negative (time:CDRSB0 ≈ -0.02), so patients who start with better cognition tend to show a faster increase in BPRS, whereas those with worse baseline cognition worsen more slowly. This answers the main question at the population level: BPRS worsens over time, and the speed of worsening is linked to baseline cognitive status.

Model Limitations

In Question 3 we use a multivariate repeated-measures model. This type of model is mainly suitable when each subject is measured at a small number of fixed time points. Our study fits this setting: BPRS is planned at seven yearly visits, and the software (REPEATED in SAS or `time | patid` in R) can handle that some of these visits are missing for a given patient.

When the number of follow-up times becomes large, multivariate regression is only workable under very specific choices for the mean and covariance; unstructured means and unstructured covariances would otherwise require too many parameters. For unbalanced data with irregular or missing visits, some standard covariance patterns (such as Toeplitz or AR(1)) are not meaningful unless time points are equally spaced, and compound symmetry would impose very strong assumptions by forcing the same correlation for all pairs of visits.

In our data, we have seven fixed annual measurements but substantial dropout. This is still a small number of time points to allow an unstructured covariance. We therefore combined a reduced mean structure with an unstructured covariance over the seven visits, which keeps the model manageable and matches the correlation pattern seen in the exploratory plots.

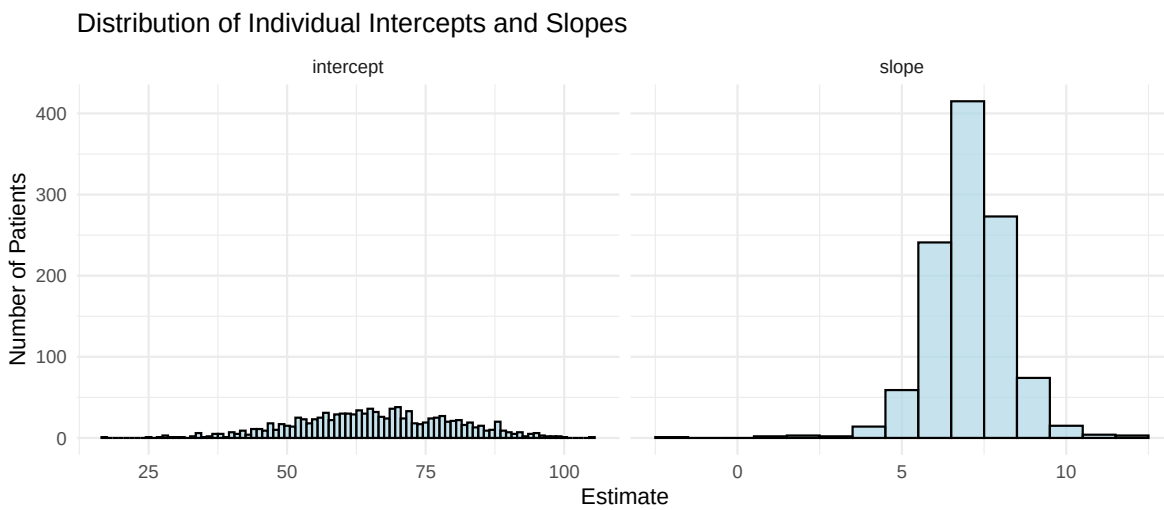
Question 4

In the first stage of a two-stage analysis, we fit a separate linear regression model for each patient to estimate their individual intercept and slope over time. For each patient i, the model is:

$$Y_{ij} = \beta_{0i} + \beta_{1i} \cdot \text{time}_j + e_{ij}$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_{0i} is the patient-specific intercept (estimated baseline BPRS).
- β_{1i} is the patient-specific slope (estimated rate of change in BPRS).
- e_{ij} are the individual error terms, assumed to be independent with mean zero.



Stage 2: Relate Intercepts and Slopes to Covariates

Model for the Intercepts

In the second stage, we model the estimated patient-specific intercepts ($\hat{\beta}_{0i}$) as a function of baseline covariates to understand which factors predict the initial BPRS score. The model is:

$$\begin{aligned}\hat{\beta}_{0i} =& \gamma_{00} + \gamma_{01} \cdot \text{sex}_i + \gamma_{02} \cdot \text{age}_i + \gamma_{03} \cdot \text{bmi}_i + \gamma_{04} \cdot \text{job}_i + \gamma_{05} \cdot \text{adl}_i \\ &+ \gamma_{06} \cdot \text{wzc}_i + \gamma_{07} \cdot \text{cdrsb0}_i + \gamma_{08} \cdot \text{abpet0}_i + \gamma_{09} \cdot \text{taupet0}_i + \epsilon_{0i}\end{aligned}$$

where:

- $\hat{\beta}_{0i}$ is the estimated intercept for patient i from Stage 1.
- γ_{00} is the average intercept for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{01} is the intercept difference for Female vs Male.
- γ_{02} is the intercept change per 1-year increase in age.

Table 3: Two-Stage Intercept Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-130.85 [-137.31, -124.39]	3.29	-39.76	<0.01
sexFemale	-0.22 [-0.51, 0.06]	0.15	-1.52	0.13
age	2.36 [2.32, 2.41]	0.02	101.88	<0.01
bmi	0.51 [0.45, 0.58]	0.03	15.07	<0.01
jobPaid Job	-18.28 [-19.25, -17.32]	0.49	-37.15	<0.01
adl	2.03 [1.90, 2.15]	0.06	31.98	<0.01
wzcNursing Home	1.54 [1.22, 1.86]	0.16	9.41	<0.01
cdrsb0	0.00 [-0.02, 0.02]	0.01	0.12	0.90
abpet0	0.35 [-0.16, 0.86]	0.26	1.35	0.18
taupet0	0.53 [-2.27, 3.33]	1.43	0.37	0.71

- γ_{03} is the intercept change per 1-unit increase in BMI.
- γ_{04} is the intercept difference for Paid Job vs No Paid Job.
- γ_{05} is the intercept change per 1-unit increase in ADL.
- γ_{06} is the intercept difference for Nursing Home vs Lives at Home.
- γ_{07} is the intercept change per 1-unit increase in baseline CDR-SB (cdrsb0).
- γ_{08} is the intercept change per 1-unit increase in baseline Amyloid PET (abpet0).
- γ_{09} is the intercept change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{0i} is the error term.

Model for the Slopes

Similarly, we model the estimated patient-specific slopes ($\hat{\beta}_{1i}$) as a function of baseline covariates to see what predicts the rate of change in BPRS. The model is:

$$\begin{aligned}\hat{\beta}_{1i} =& \gamma_{10} + \gamma_{11} \cdot \text{sex}_i + \gamma_{12} \cdot \text{age}_i + \gamma_{13} \cdot \text{bmi}_i + \gamma_{14} \cdot \text{job}_i + \gamma_{15} \cdot \text{adl}_i \\ &+ \gamma_{16} \cdot \text{wzc}_i + \gamma_{17} \cdot \text{cdrsb0}_i + \gamma_{18} \cdot \text{abpet0}_i + \gamma_{19} \cdot \text{taupet0}_i + \epsilon_{1i}\end{aligned}$$

where:

- $\hat{\beta}_{1i}$ is the estimated slope for patient i from Stage 1.
- γ_{10} is the average slope for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{11} is the slope difference for Female vs Male.
- γ_{12} is the slope change per 1-year increase in age.

Table 4: Two-Stage Slope Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	6.37 [2.97, 9.77]	1.73	3.68	<0.01
sexFemale	-0.04 [-0.19, 0.11]	0.08	-0.55	0.58
age	0.01 [-0.01, 0.04]	0.01	1.02	0.31
bmi	0.00 [-0.03, 0.04]	0.02	0.09	0.93
jobPaid Job	-0.34 [-0.85, 0.17]	0.26	-1.32	0.19
adl	0.05 [-0.02, 0.11]	0.03	1.37	0.17
wzcNursing Home	0.19 [0.02, 0.36]	0.09	2.23	0.03
cdrsb0	-0.02 [-0.03, -0.01]	0.01	-4.30	<0.01
abpet0	-0.07 [-0.34, 0.20]	0.14	-0.53	0.60
taupet0	-0.13 [-1.61, 1.34]	0.75	-0.17	0.86

- γ_{13} is the slope change per 1-unit increase in BMI.
- γ_{14} is the slope difference for Paid Job vs No Paid Job.
- γ_{15} is the slope change per 1-unit increase in ADL.
- γ_{16} is the slope difference for Nursing Home vs Lives at Home.
- γ_{17} is the slope change per 1-unit increase in baseline CDR-SB (cdrsb0).
- γ_{18} is the slope change per 1-unit increase in baseline Amyloid PET (abpet0).
- γ_{19} is the slope change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{1i} is the error term.

Question 5

Plausible random effects model

To understand how psychiatric symptoms evolve over time in Alzheimer's disease, we developed a statistical model that accounts for repeated measurements in each patient. This model allows each patient to have their own starting level of symptoms and their own rate of change over time. The outcome of interest was the Brief Psychiatric Rating Scale (BPRS), measured annually over six years.

We model the longitudinal BPRS scores using a linear mixed-effects model with random effects at both the patient and trial level:

$$\begin{aligned}\text{BPRS}_{it} = (\beta_0 + b_{0i} + u_{0j}) + (\beta_1 + b_{1i})\,\text{time}_{it} + \beta_2\text{sex}_i + \beta_3\text{age}_i + \beta_4\text{bmi}_i \\ +\beta_5\text{job}_i + \beta_6\text{adl}_i + \beta_7\text{wzc}_i + \beta_8\text{cdrsb0}_i + \beta_9(\text{time}_{it} \cdot \text{cdrsb0}_i) + \epsilon_{it}\end{aligned}$$

where:

- BPRS_{it} is the BPRS score for patient i in trial j at time t.
- $\beta_0, \dots, \beta_9$ are the fixed (population-level) effects.
- b_{0i} is the patient-specific random intercept.
- b_{1i} is the patient-specific random slope for time.
- u_{0j} is the trial-specific random intercept.
- ϵ_{it} is the residual error term.

The patient-level random effects are assumed to follow a bivariate normal distribution:

$$\begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{b0}^2 & \sigma_{b0,b1} \\ \sigma_{b0,b1} & \sigma_{b1}^2 \end{pmatrix} \right),$$

and the trial-level random intercept follows:

$$u_{0j} \sim N(0, \sigma_{\text{trial}}^2).$$

Model Motivation and Justification

Here, We chose to fit linear mixed effects model because our dataset contains repeated BPRS measurements for the same patients over 6 years and across multiple clinical trials, with clear variability both in baseline symptom severity and rate of change between patients. An unstructured covariance matrix was used for the random effects, allowing the intercept and slope to be correlated. The model included:

1) Random intercepts and random slopes for each patient, reflecting that individuals differ both in their baseline psychiatric symptom severity and their rate of change over time.
2) A random intercept for trial was added to account for systematic differences across the 25 clinical trials.
3) Fixed effects included all clinically relevant baseline covariates (age, sex, BMI, job status, living situation, CDR-SB), ensuring adjustment for major patient characteristics.
4) An interaction term between time and CDRSB was included to assess whether cognitive impairment at baseline affects the rate of psychiatric symptom progression.

Estimation Method

The model was fitted using Restricted Maximum Likelihood (REML), which provides more reliable and less biased estimates of variance components, especially because our dataset had many clusters (patients, trials).Because our model includes random intercepts for both patients and trials, REML is appropriate to ensure stable estimates of these random effect variances.

Assumptions made

1) Linearity: BPRS changes linearly over the follow-up period.
2) Normality: Random effects and residual errors are approximately normally distributed.
3) Independence: After accounting for random effects, residuals are independent.
4) Homoscedasticity: The residual errors have constant variance across time and subjects.
5) Missing data (dropout) are at least missing at random (MAR) conditional on the covariates and random effects.
6) Correct Covariance Structure: The specified random-effects structure (random intercepts and slopes) appropriately captures within-subject and within-trial correlation.

Limitations

1) More complex to interpret for non-statisticians.The presence of fixed effects, random effects, and correlation structures makes interpretation harder compared to ordinary regression
2) Results depend on correct specification of random-effects structure. Misspecifying random intercepts/slopes or the covariance structure can lead to biased estimates or incorrect standard errors.
3) Convergence issues can arise with complex models. Models with multiple random effects, unstructured covariance matrices, or small sample sizes may fail to converge or produce unreliable estimates.
4) Assumptions such as normality of random effects and linearity may not fully hold in real data. Although LME models are robust, violation of assumptions especially the normality of random effects can affect estimates, particularly with small sample sizes.
5) MAR assumption cannot be fully verified.The model assumes missing data are Missing At Random, but this assumption is untestable and can lead to biased results if dropout relates to unobserved outcomes.

Table 5: Mixed Model with Random Intercept and Slope Results

effect	term	estimate	std.error	statistic	p.value

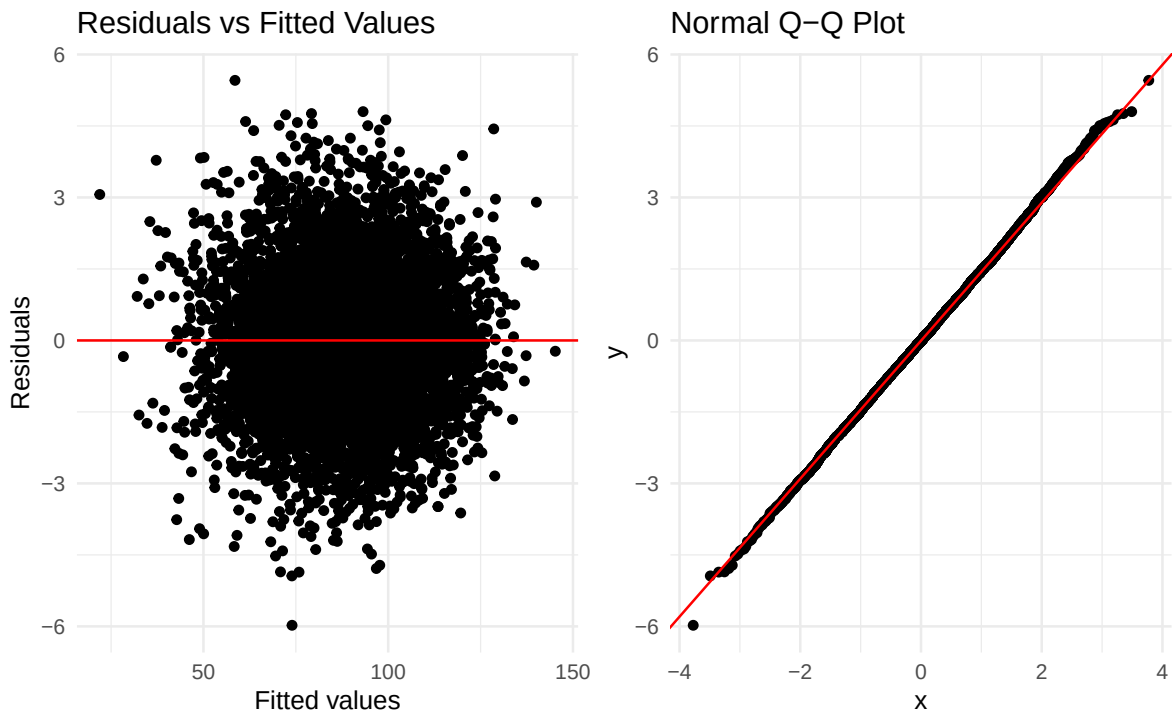
fixed	(Intercept)	-60.29 [-68.48, -52.11]	4.17	-14.46	<0.01
fixed	time	7.22 [7.14, 7.30]	0.04	179.32	<0.01
fixed	sexFemale	-0.16 [-0.33, 0.01]	0.09	-1.84	0.07
fixed	age	1.70 [1.62, 1.78]	0.04	42.36	<0.01
fixed	bmi	0.18 [0.12, 0.23]	0.03	6.54	<0.01
fixed	jobPaid Job	-5.14 [-6.71, -3.56]	0.80	-6.39	<0.01
fixed	adl	0.07 [-0.17, 0.30]	0.12	0.55	0.58
fixed	wzcNursing Home	1.92 [1.72, 2.11]	0.10	19.56	<0.01
fixed	cdrsb0	0.00 [-0.01, 0.02]	0.01	0.26	0.79
fixed	time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.25	<0.01

Diagnostics: Evaluating Model Appropriateness

To ensure that the linear mixed-effects model is statistically reliable, several diagnostic checks were performed. These checks help determine whether the model's assumptions are met, and whether the results are reliable.

Normality of Residuals (Q–Q Plot)

Figure 8 (right plot) presents the Normal Q–Q plot which compares the distribution of the model residuals (errors) to what would be expected if they followed a normal distribution, which is an underlying assumption of linear mixed-effects models.The residuals aligned closely with the expected diagonal line, showing two deviations at each tail.The assumption of normally distributed residuals is largely satisfied. This supports the reliability of the estimated effects.



Residuals vs. Fitted Values Plot

Figure 8(right) plot assesses whether residuals are randomly distributed around zero, which reflects the assumption of constant variability ("homoscedasticity").Residuals are spread evenly around zero with no funnel shape or curvature. This indicates Variance is stable across time, the model does not systematically under or over predict symptoms and the linearity assumption is appropriate. The model fits the data well and the variability is appropriately captured.

Distribution of Patient-Specific Intercepts and Slopes

To evaluate whether the random-effects model appropriately captured differences between patients, we examined the distribution of the estimated subject-specific intercepts (patients' baseline BPRS scores) and subject-specific slopes (their rates of symptom change over time)(Figure 9). The random intercepts(Figure 9: left Q-Q plot) follow the theoretical normal line very closely, with only minor deviations at the extreme upper and lower tails. This suggests that the normality assumption for the random intercepts is well satisfied. The distribution is symmetric and shows no major outliers, indicating that patient-level baseline differences in BPRS are reasonably captured by a normal random intercept.The random slopes(Figure 9:Middle Q-Q plot) also align closely with the diagonal reference line, again with only slight deviations in the two tails. This indicates that the random slopes are approximately normally distributed, supporting the assumption that individual differences in rate of change follow a normal pattern.

No strong violations or heavy-tailed behavior are present. The scatterplot of random intercepts versus random slopes (Figure 9: Scatterplot) shows cloud of points with no strong linear relationship. This implies that patients' baseline BPRS levels are not strongly associated with their rate of change over time. In other words, starting severity does not systematically predict whether a patient improves faster or slower. The spread is balanced around zero for both intercepts and slopes, confirming that the random-effects structure is behaving as expected.

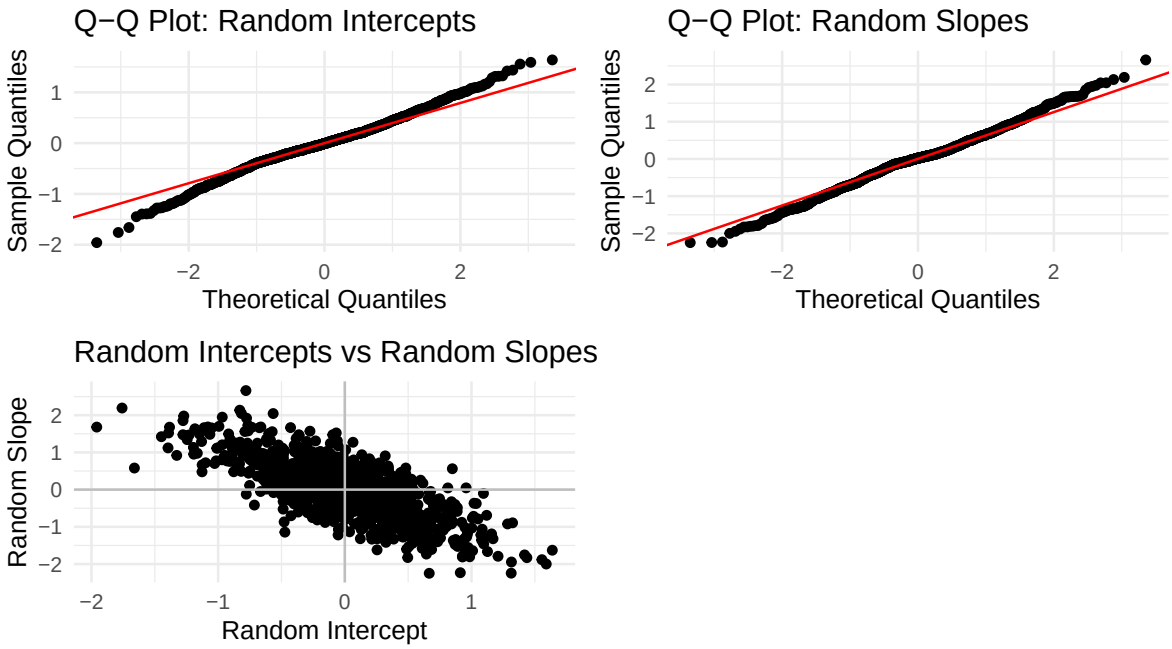
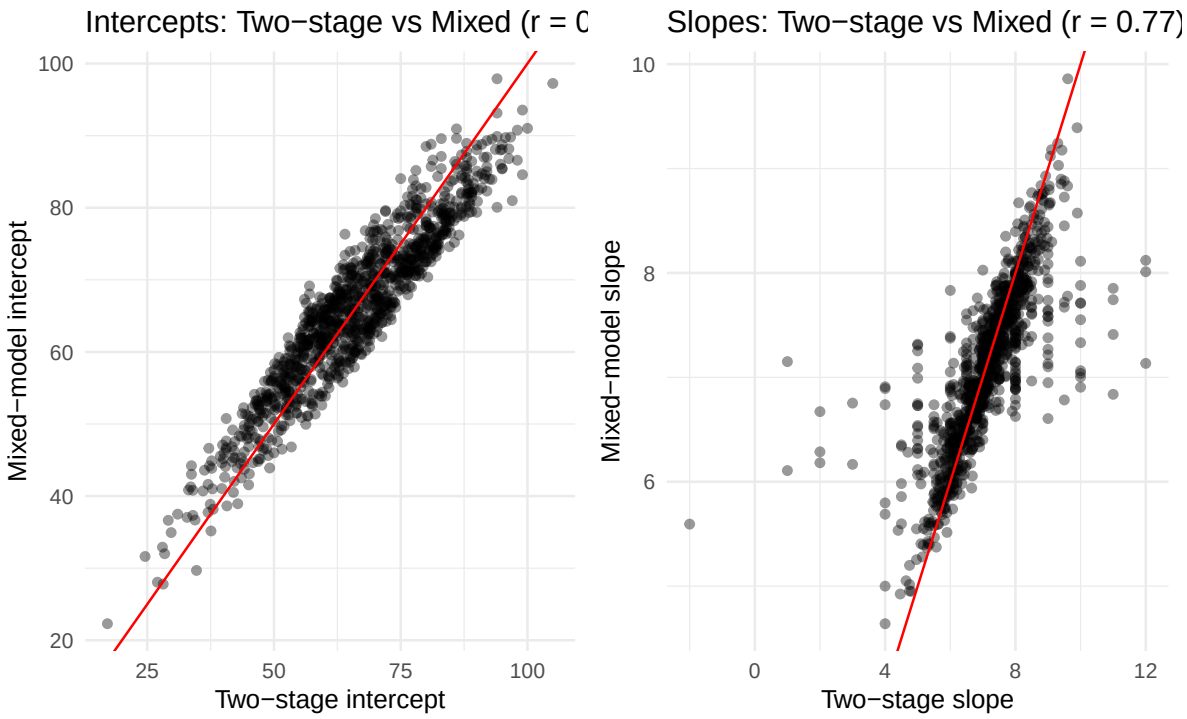


Figure 9: Diagnostics of Random Effects:
Q−Q plots for random intercepts (left) and random slopes (middle), and their joint distribution (right).

Comparison With the Two-Stage Analysis

To compare the two-stage approach with the mixed-effects model, we examined patient-specific intercepts and slopes estimated by each method. The intercepts showed very strong agreement between the two approaches (Figure 10, left), with a correlation of r = 0.94 and points closely following the 45-degree line. This indicates that baseline BPRS levels are estimated consistently in both methods. In contrast, the agreement for slopes was moderate (Figure 10, right; r = 0.77), and the two-stage estimates displayed greater variability and several extreme values. This pattern reflects the inherent instability of estimating individual change rates from limited per-patient data. The mixed-effects model, by using all availabledata simultaneously, shrinks extreme slopes toward the population mean and produces more stable and reliable estimates. Overall, the mixed model provides a clearer and less noisy picture of individual symptom trajectories, particularly for the rate of change over time.

Overall, the two methods give broadly similar patient-specific results, but the mixed model provides a clearer and more stable representation of individual symptom trajectories, particularly for the rate of change over time



Why are the mixed-model estimates still preferred?

Compared to the two-stage approach, mixed-effects models are preferred because they correctly account for the correlation among repeated measurements within each patient, whereas the two-stage method treats each patient's data independently. Mixed models also allow a flexible covariance structure for intercepts and slopes, leading to more accurate uncertainty estimates. In contrast, the two-stage approach produces unstable slopes when patients have few observations and cannot model within-patient correlation. Overall, the mixed model provides more reliable and statistically valid patient-specific estimates.

Results

Variance components from mixed model

The variance components show that, besides differences between patients, there is a big between-centre (trial) variance, with trial-level $SD \approx 2.8$, much larger than the patient-level SD. This means average outcomes differ substantially across centres.

Table 6: Random Effects: Variance Components from Mixed Model

	Groups	Name	Variance	Std.Dev.	Corr
(Intercept)	patid	(Intercept)	0.807	0.898	NA
time	patid	time	0.686	0.828	-0.611
(Intercept)1	trial	(Intercept)	8.004	2.829	NA
1	Residual		2.698	1.643	NA

The linear mixed-effects model was fitted to 6,220 observations from 1,253 patients across 25 trials. The REML criterion at convergence was 27,121.3, and scaled residuals ranged from -3.64 to 3.32.The random-effects structure included patient-level random intercepts and random slopes for time. These patient-specific intercepts and slopes were correlated. The model also included trial-level random intercepts to account for between-trial variation. The full set of estimated variances, standard deviations, and correlations for all random-effect terms is presented in Table 5.

Several fixed effects were statistically significant, including the intercept, time, age, BMI, job status, WZC category, and the time-by-cdrsb0 interaction, all with ($p<0.001$). The effects of sex, ADL, and the main effect of cdrsb0 were not statistically significant. Full estimates, standard errors, and significance levels for all fixed effects are presented in Table 4.

Discussion

Using a mixed-effects model, we examined how psychiatric symptoms change over time in patients with Alzheimer's disease. This modelling approach accounts for individual differences in both baseline symptom severity and rates of progression.

The analysis showed that psychiatric symptoms worsen steadily over time, with patients on average increasing by about 7 BPRS points per year. However, not all patients start at the same level or decline at the same speed. Our model captured this variation and revealed important patient characteristics that help explain these differences.

Patients who were older, had higher BMI, poorer daily functioning (ADL), lived in a nursing home, or did not have a paid job tended to have more severe psychiatric symptoms at baseline. In addition, patients with more impaired cognition at diagnosis (higher CDR-SB) showed faster worsening of psychiatric symptoms over time.

Overall, this approach provides a detailed picture of how psychiatric symptoms evolve in Alzheimer's disease. It shows which baseline factors are linked with faster decline, and it gives individualized estimates of symptom progression. These findings can help to better understand

patient variability, identify patients who may be at higher risk for rapid deterioration, and guide monitoring and care strategies over the course of the disease.

Reflection on parameterization

The chosen parameterization had two main components:

Mean structure (fixed effects): We specified a linear effect of time, adjusted for sex, age, BMI, job status, ADL, nursing home residence (WZC), baseline CDR-SB, amyloid PET, and the interaction between time and CDR-SB. This reflects a clinical hypothesis that:

1) symptom severity depends on demographic, functional, and disease severity markers at baseline, and
2) cognitive status modifies the trajectory of psychiatric symptoms (time $\times$ CDRSB0).

Although the assumption of linear change simplifies interpretation, non-linear patterns could be explored in future work if justified..

For the covariance structure, we selected patient-level random intercepts and random slopes for time with an unstructured covariance matrix. This parameterization captures individual differences in baseline severity and rates of change while allowing them to be correlated. Trial-level random intercepts accounted for between-trial variation. Overall, this structure provides a good balance between flexibility and interpretability. Alternative parameterizations, such as non-linear time or more complex random effects, could be considered, but the chosen model is appropriate and coherent for the current analysis.

Recommendations for future similar studies

Based on what we observed in the exploratory plots and models, we suggest the following design and data collection improvements for future work on BPRS in Alzheimer patients.

1. Address informative dropout and missingness

- Patients with higher baseline BPRS missed more follow-up visits, and overall missingness rose above 40% by Year 6. This shows that dropout is related to disease severity, and is unlikely to be completely random. - At enrolment, consider whether including patients with extremely severe psychiatric symptoms at baseline will lead to very early dropout. If this subgroup is retained, plan in advance how their missing data will be handled in the analysis. - Record reasons for dropout in a structured way (death, institutionalisation, moving away, withdrawal of consent, adverse events). This will help future analyses distinguish between random and informative missingness and allow sensitivity analyses.

2. Standardise clinical site procedures

- The LOESS plots by clinical centre showed clear differences in BPRS trajectories between centres. In the mixed-effects model, a random intercept for trial was needed to account for important between-centre variation.
  - Tighten and document SOPs for BPRS assessment across centres, including rater training and calibration, to reduce artificial differences due to measurement.
  - If centres serve populations with different baseline severity, consider stratifying enrolment by baseline BPRS or CDR-SB so that severity is more evenly distributed over centres and treatment groups.

3. Collect richer covariate information

- In the two-stage analysis and mixed-effects model, there was still substantial variation in patient-specific intercepts and slopes after adjusting for age, sex, BMI, ADL, and cognition. This suggests important patient differences in progression that are not yet explained.
  - In future studies, collect additional longitudinal information on treatment and care: medication classes and doses, psychosocial or behavioural interventions, caregiver burden, and comorbidities (e.g. depression, cardiovascular disease). These variables may help explain which patients deteriorate faster and could identify modifiable risk factors.

4. Improve biomarker data collection

- Tau PET (taupet0) showed almost no variability from about Year 3 onwards, which made it uninformative for explaining changes in BPRS. This limits its use in longitudinal models.
  - Review and standardise biomarker measurement protocols across centres to avoid ceiling or floor effects.
  - Where possible, broaden the range of biomarkers collected, and ensure they are measured with enough precision and variability over time to be useful in explaining clinical trajectories.

Appendix

Table 7: Gls Full Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-129.58 [-133.00, -126.15]	1.75	-74.20	<0.01
time	5.96 [4.27, 7.64]	0.86	6.91	<0.01
eduLower secondary	0.20 [-0.13, 0.52]	0.17	1.19	0.23
eduUpper secondary	0.12 [-0.19, 0.42]	0.16	0.75	0.46
eduHigher	0.19 [-0.11, 0.50]	0.16	1.23	0.22
sexFemale	-0.24 [-0.49, 0.00]	0.13	-1.95	0.05
age	2.36 [2.32, 2.40]	0.02	126.37	<0.01
bmi	0.51 [0.45, 0.57]	0.03	17.00	<0.01
jobPaid Job	-18.13 [-18.91, -17.35]	0.40	-45.69	<0.01
adl	2.00 [1.90, 2.10]	0.05	38.57	<0.01
wzcNursing Home	1.63 [1.35, 1.92]	0.15	11.16	<0.01
cdrsb0	0.00 [-0.01, 0.02]	0.01	0.21	0.83
abpet0	0.34 [-0.10, 0.78]	0.22	1.52	0.13
time:sexFemale	-0.03 [-0.16, 0.09]	0.06	-0.52	0.61
time:age	0.01 [-0.00, 0.03]	0.01	1.59	0.11
time:bmi	-0.00 [-0.03, 0.03]	0.01	-0.06	0.95
time:jobPaid Job	-0.39 [-0.78, 0.01]	0.20	-1.90	0.06
time:adl	0.05 [-0.00, 0.11]	0.03	1.95	0.05
time:wzcNursing Home	0.14 [-0.00, 0.29]	0.07	1.90	0.06
time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.10	<0.01
time:abpet0	-0.07 [-0.29, 0.16]	0.11	-0.58	0.56